



南京邮电大学
Nanjing University of Posts and Telecommunications

科学方法写作汇报

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Face reconstruction using Principal Component Analysis



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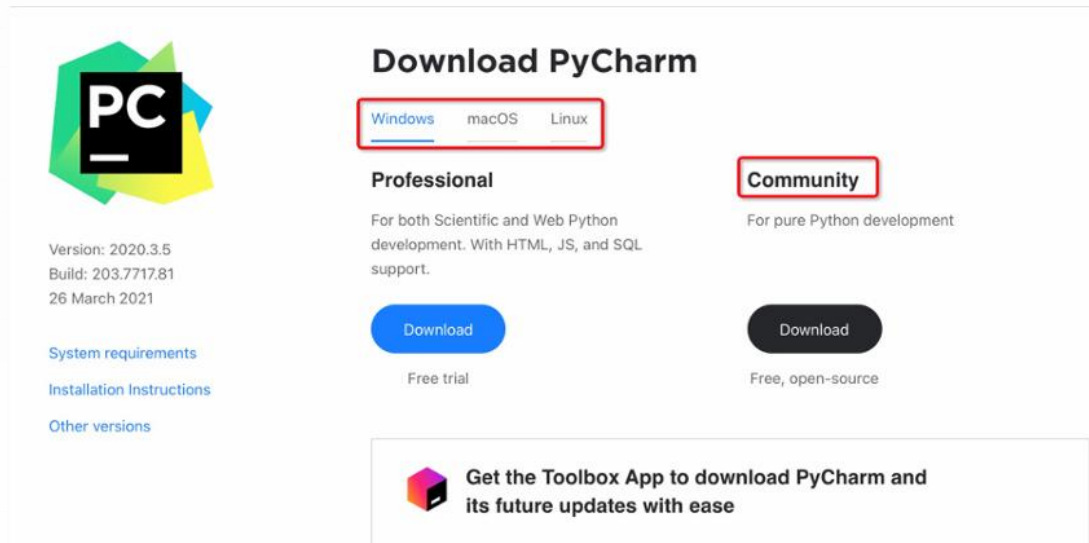
Introduction to PCA

- Reduce dimensions while keeping key information
- Identify directions with most variance
- Remove noise, keep important patterns
- Easier visualization of high-dimensional data
- Apply to images: face recognition and reconstruction

Face reconstruction using Principal Component Analysis



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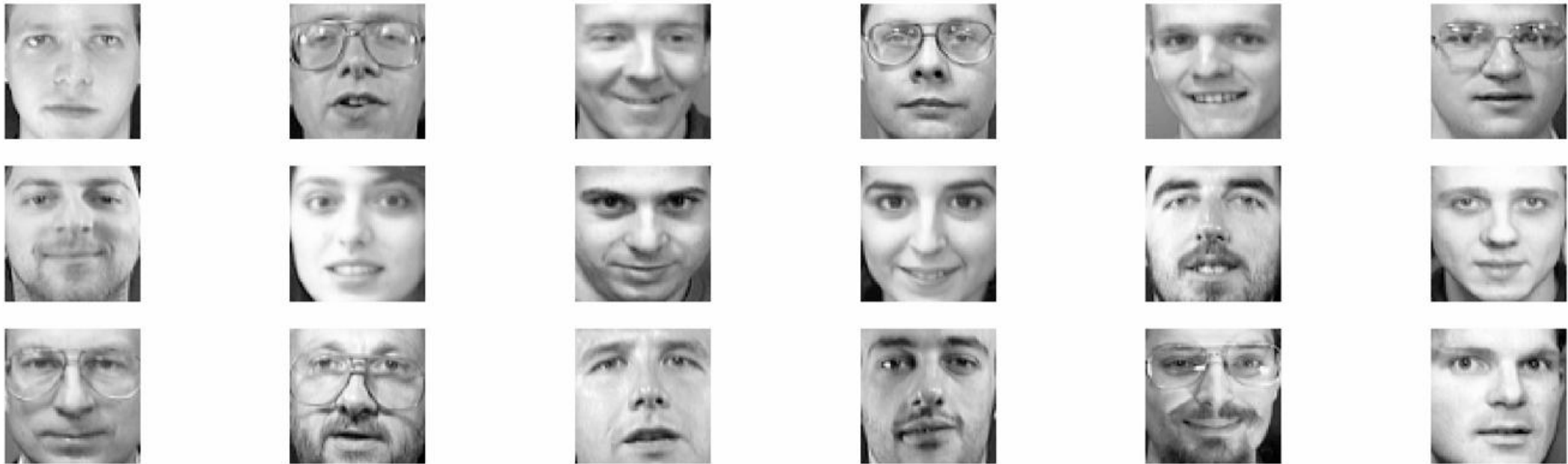


Experimental Environment

- Programming Language: Python 3.10
- Platform: PyCharm
- Libraries: NumPy, Pandas, Matplotlib, scikit-learn
- Operating System: Windows 10

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.datasets import fetch_olivetti_faces
from sklearn.decomposition import PCA
from sklearn.metrics import mean_squared_error
import os, time
```

Dataset Introduction



Olivetti Faces Dataset: 400 grayscale face images (64×64) from 40 people, with varied expressions and lighting — widely used for PCA and face recognition.

Original Work

- Implemented PCA function for dimensionality reduction and feature extraction
- Selected sample faces and visualized using `matshow()`
- Reduced face data to 20-dimensional feature space
- Visualized top 20 eigenfaces
- Randomly selected a face for reconstruction
- Plotted the reconstruction process

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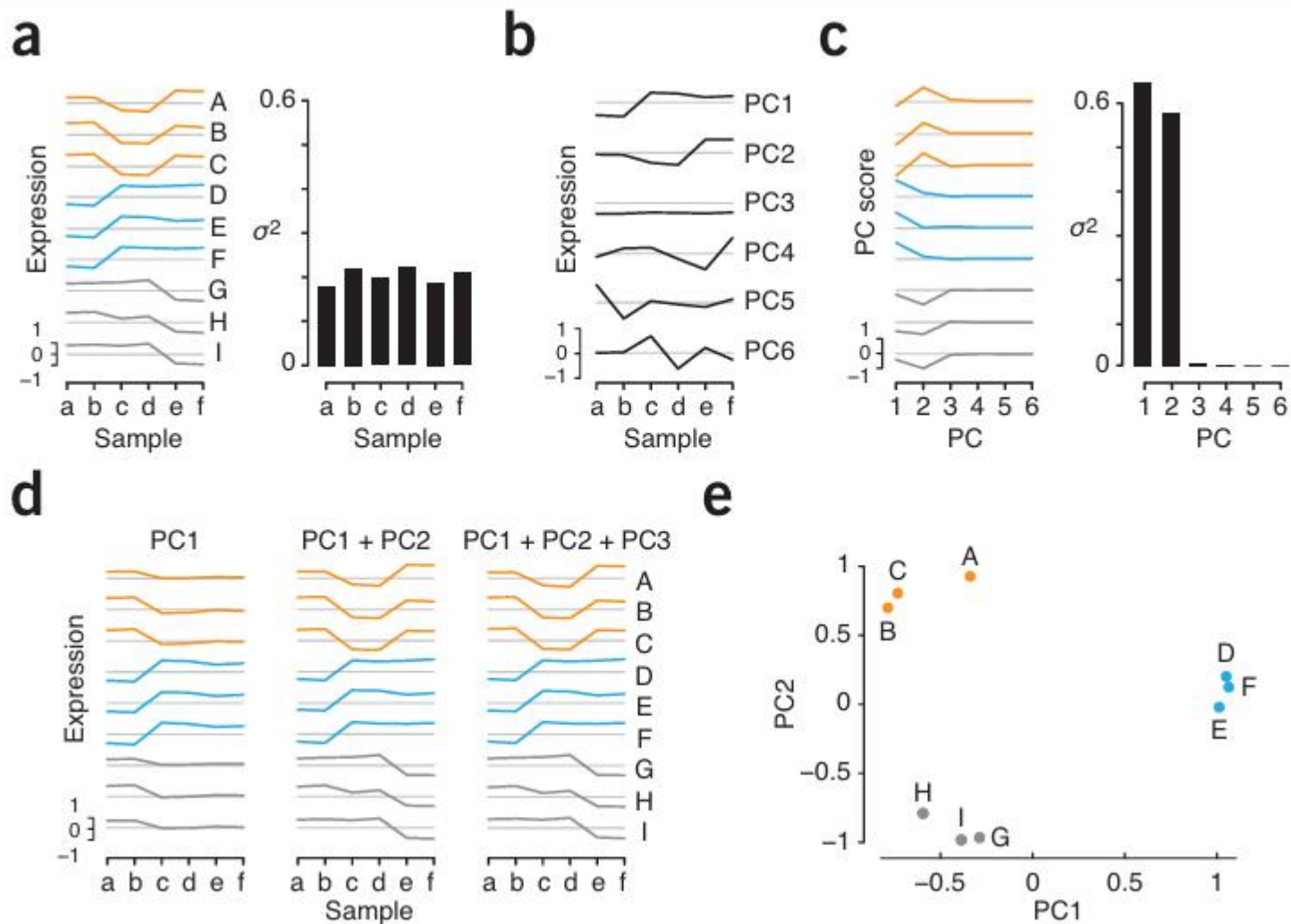


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Display Original and Feature Faces



PCA Fundamentals: A Conceptual Overview



1. Dimensionality Reduction (Fig a-c)

- **Orthogonal Extraction:** Top PCs concentrate variance —> Extreme compression & Denoising.

2. High-Fidelity Reconstruction (Fig d)

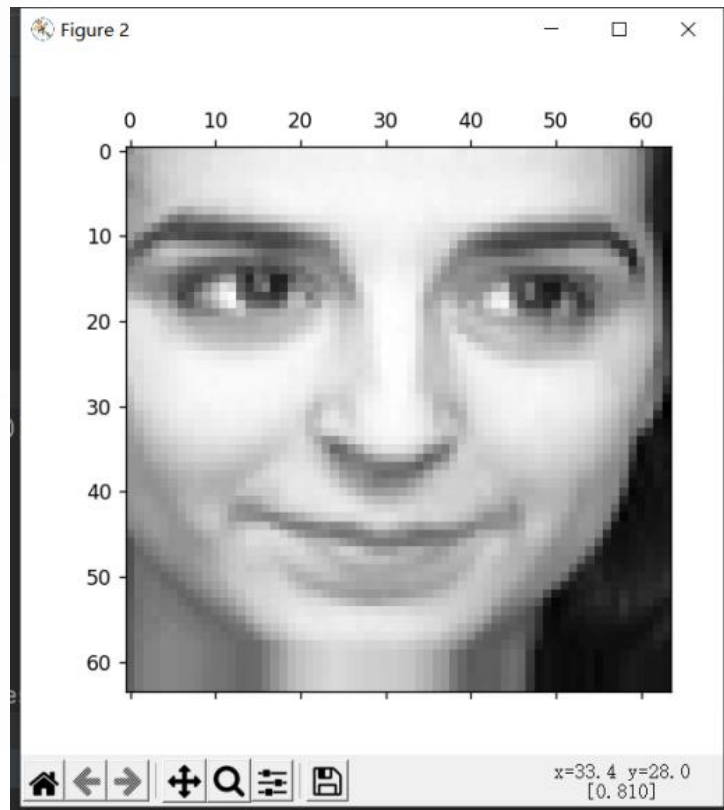
- **Inverse Transform:** Minimal core PCs (PC1 + PC2) —> Accurate detail recovery.

3. Latent Space Clustering (Fig e)

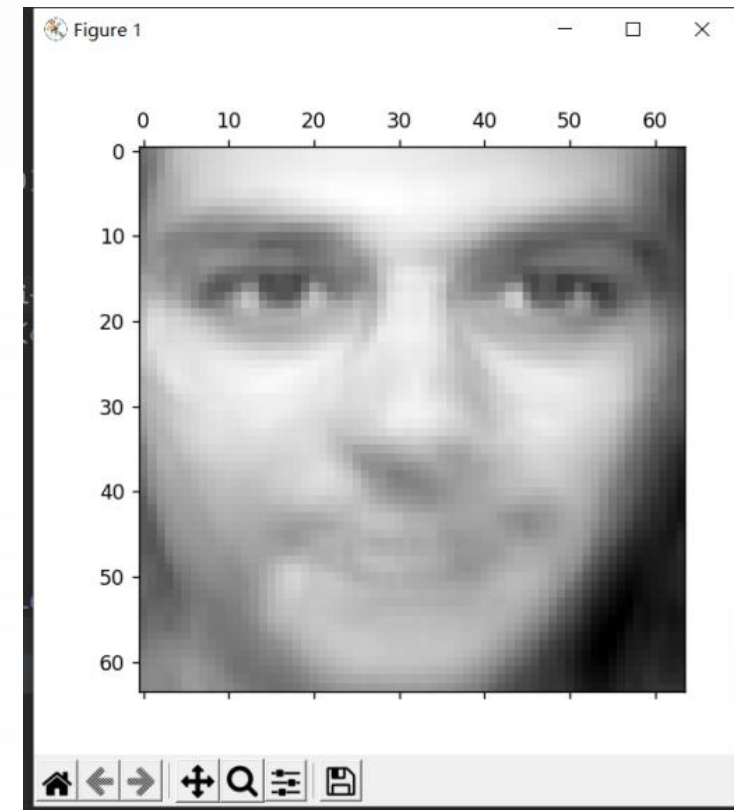
- **Low-Dim Projection:** Maps features —> High intra-class cohesion & inter-class separation.

Reduce Face Data to 20 Dimensions using PCA

```
# PCA提取特征  
faces_reduced, W, lambdas = principal_component_analysis(faces.data, 20)
```



Original face image



Reconstructed face image

Plot the Reconstruction Process of a Sample Face



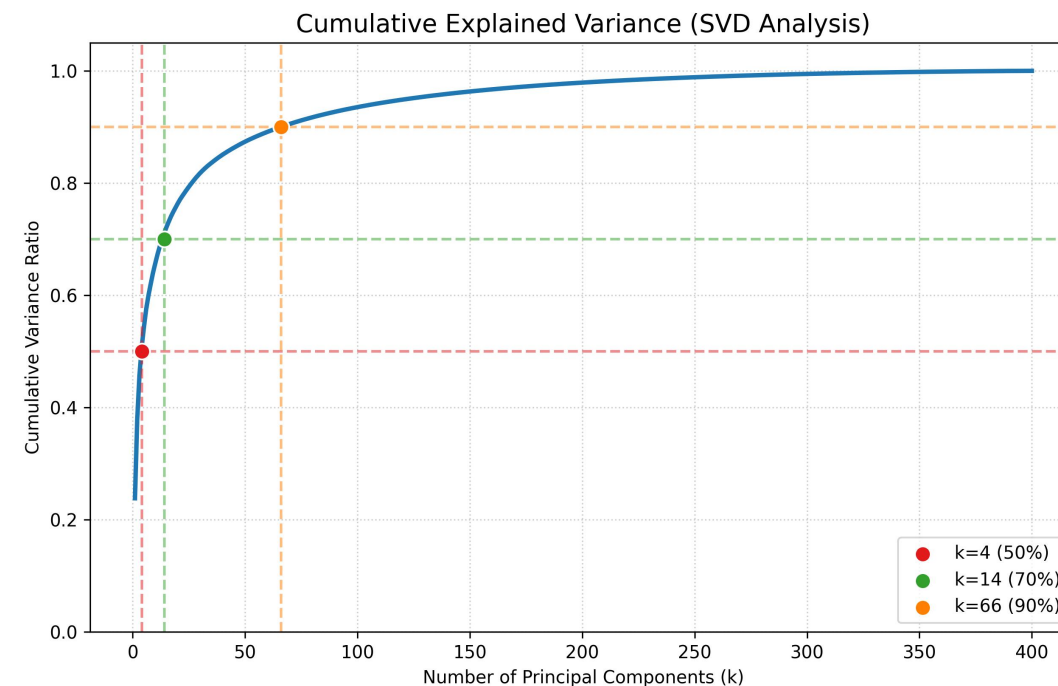
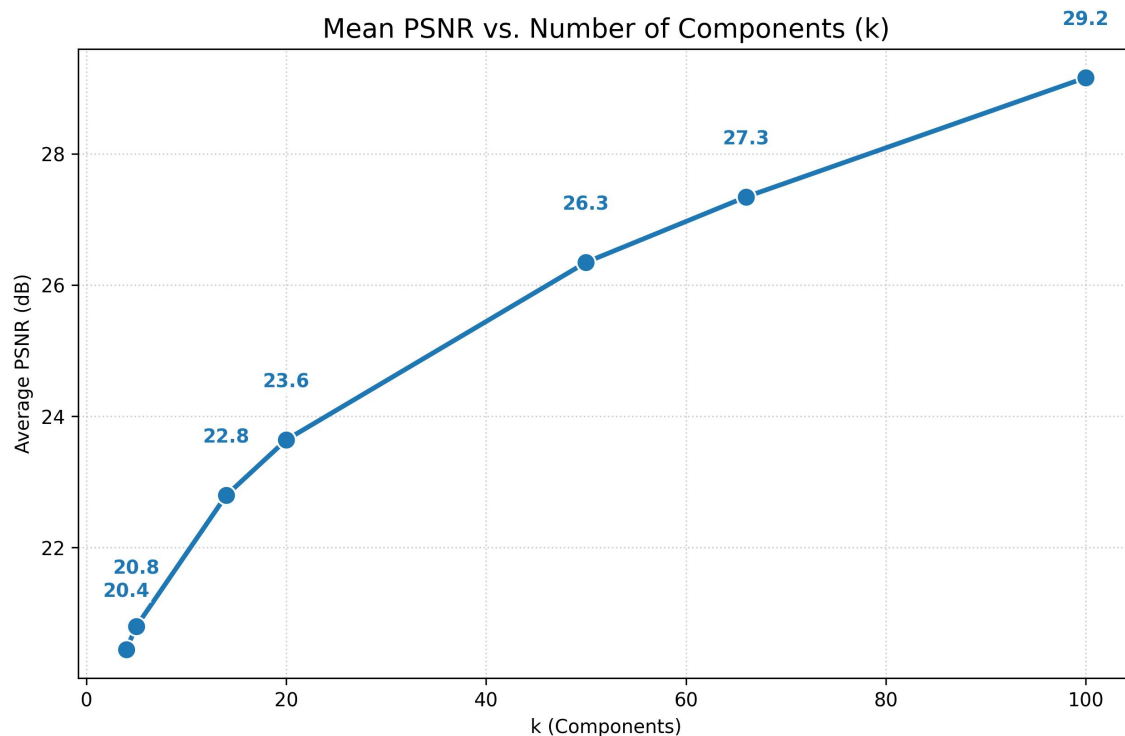
Discussion

- Limited dataset dimensionality and sample size → small number of principal components (k).
- Reconstructed faces have low visual quality and appear blurred.
- PCA parameters and quantitative metrics are not directly visible → reconstruction process is less interpretable.
- These factors reduce the precision and interpretability of face reconstruction.

Improvements and Extensions

- Dataset: Olivetti Faces, 400 images of 64×64 pixels $\rightarrow$ 4096-dimensional vectors
- Dimensionality reduction: **SVD-based PCA**^{[1][2]}
- Performance tracking: Computation time recorded for different k using time
- Reconstruction quality: **MSE, RMSE, PSNR**, used to observe and save results^[3]
- Optimized **explained variance**, **explained variance ratio** and **cumulative explained variance**^[4] $\rightarrow$ enhanced feature representation ability
- Multiple experiments: Reconstruction quality of face images under different K values. (**$K=5, 20, 50, 100$**)

SVD-based PCA



Data Compression: 90% variance $k=66$

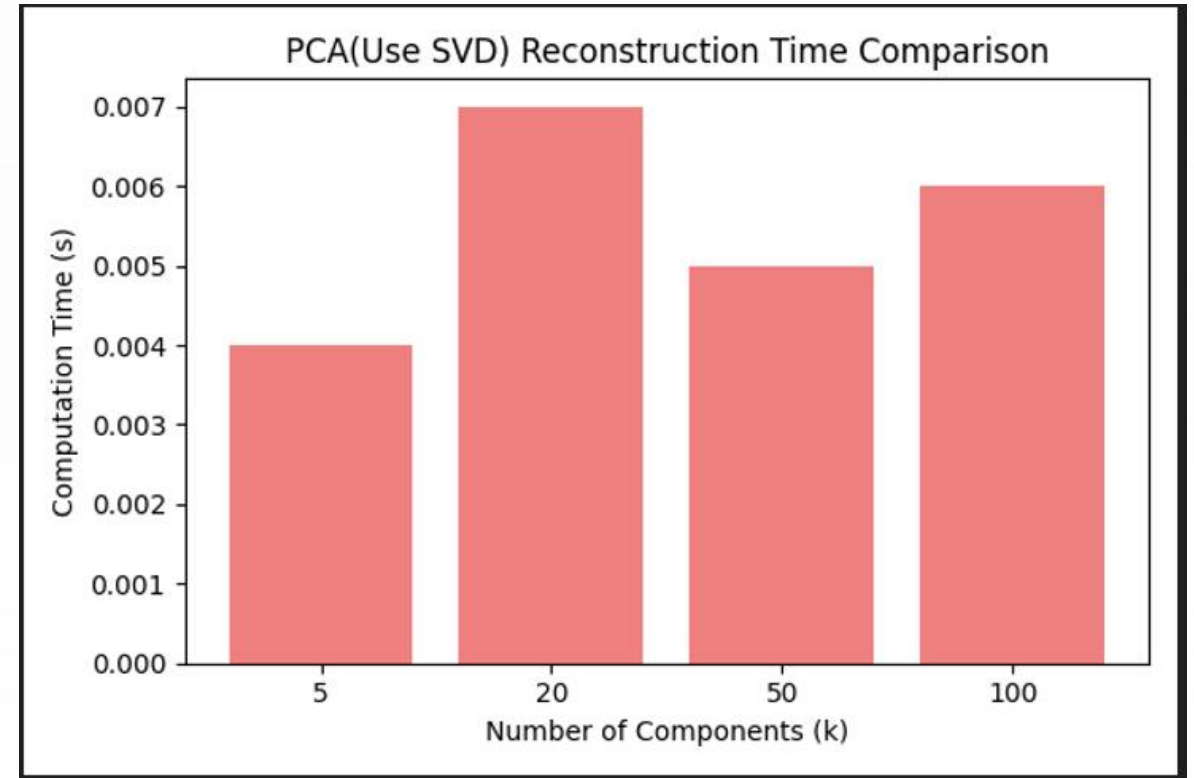
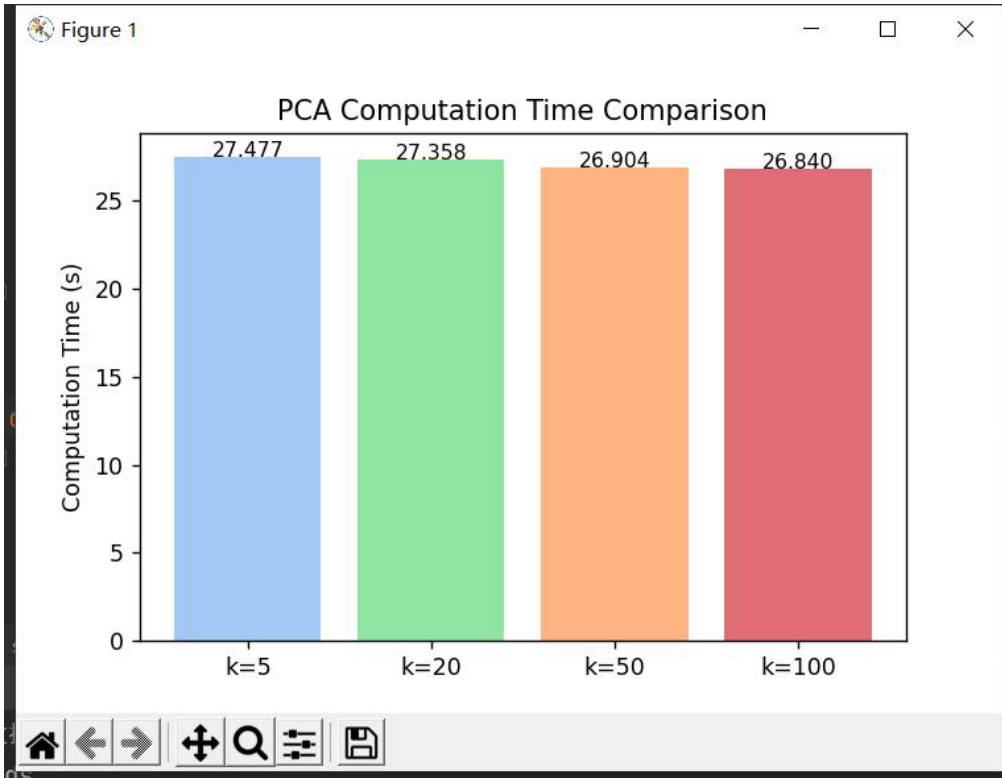
Image Quality: 27.3 dB PSNR $k=66$

Optimal Balance: $k=66$ (The Sweet Spot)

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```
F:\geek\python.exe F:/geek/python3/大数据分析/main4.py
PCA with k=5 finished in 27.4774 seconds
PCA with k=20 finished in 27.3584 seconds
PCA with k=50 finished in 26.9035 seconds
PCA with k=100 finished in 26.8404 seconds
```

进程已结束,退出代码0

Computation time recorded for different k using time

MSE, RMSE, PSNR used to observe and save results

```
F:\geek\python.exe F:/geek/python3/大数据分析/main3.py
Data shape: (400, 4096)
k=5: time=0.005s, MSE=0.00879, RMSE=0.09250, PSNR=20.80dB
k=20: time=0.005s, MSE=0.00456, RMSE=0.06664, PSNR=23.64dB
k=50: time=0.005s, MSE=0.00243, RMSE=0.04869, PSNR=26.36dB
k=100: time=0.006s, MSE=0.00125, RMSE=0.03511, PSNR=29.16dB
```

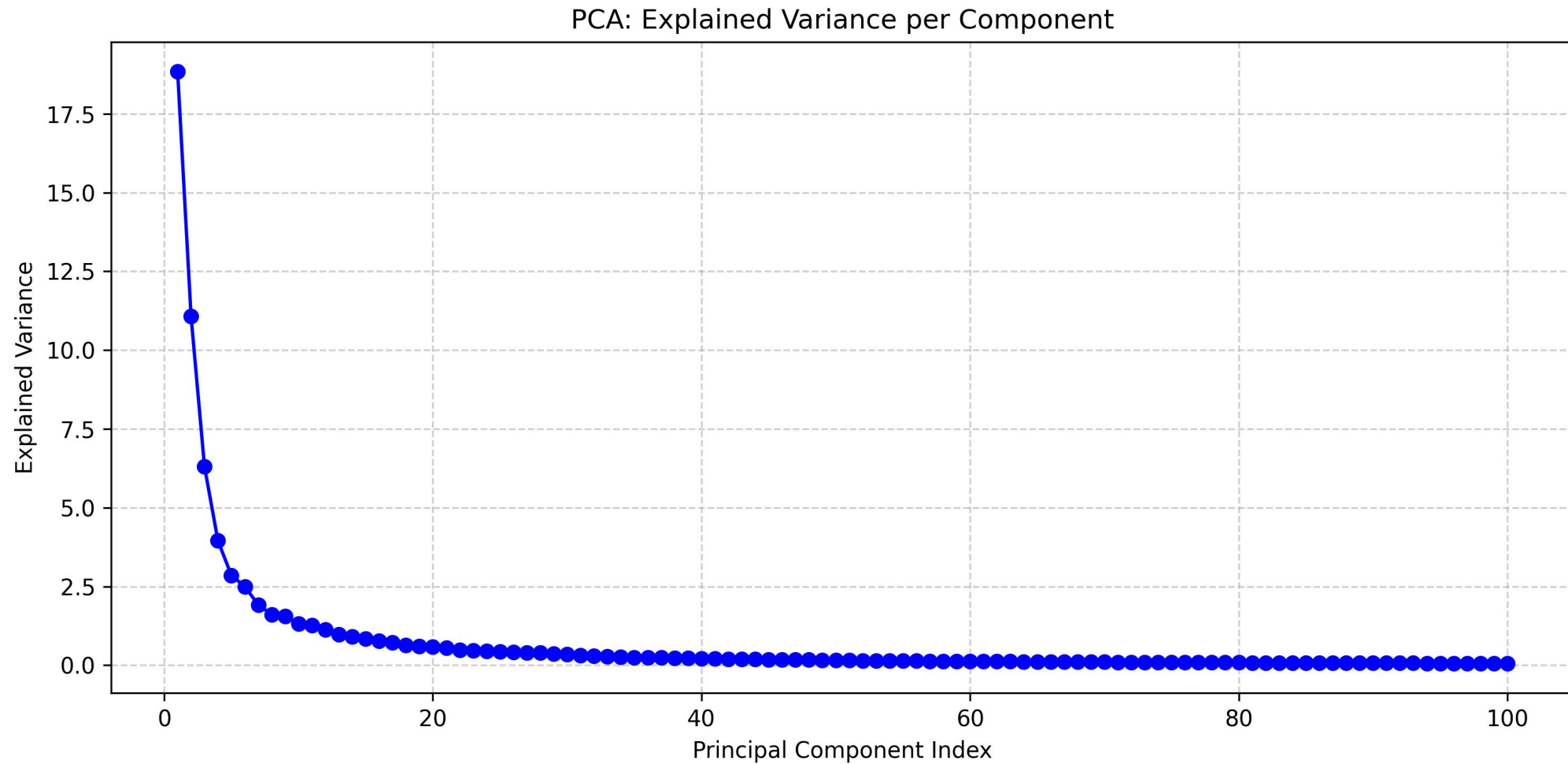
```
All results (MSE, RMSE, PSNR, Time) saved in folder: pca_results_final
```

	A	B	C	D	E
1	sample	k	mse	rmse	psnr
2	0	5	0.008238635957241058	0.09076692909002304	20.841446870020455
3	1	5	0.010809529572725296	0.10396888852119446	19.66193205992582
4	2	5	0.012289173901081085	0.1108565479516983	19.10477310137166
5	3	5	0.0052978405728936195	0.07278626412153244	22.759011149995928
6	4	5	0.00911138765513897	0.09545359015464783	20.404154753787314
7	5	5	0.012466182932257652	0.11165206134319305	19.042665044384734
8	6	5	0.006525784730911255	0.08078233152627945	21.853672565854335
9	7	5	0.007188916672021151	0.08478748053312302	21.43336550341885
10	8	5	0.008947821334004402	0.09459292143583298	20.482826962820088
11	9	5	0.013336077332496643	0.11548193544149399	18.74971894810286
12	10	5	0.009272824972867966	0.09629550576210022	20.327879375679444
13	11	5	0.00959592405706644	0.09795878827571869	20.179131977329163
14	12	5	0.011664372868835926	0.1080017238855362	19.331386059422833
15	13	5	0.01108098216354847	0.10526624321937561	19.554217441829227
16	14	5	0.01103438250720501	0.10504467040300369	19.57251965285919
17	15	5	0.010722854174673557	0.1035512164235115	19.696896001538402
18	16	5	0.011561947874724865	0.107526496052742	19.369689929067942
19	17	5	0.010499360039830208	0.10246638208627701	19.788371713720124

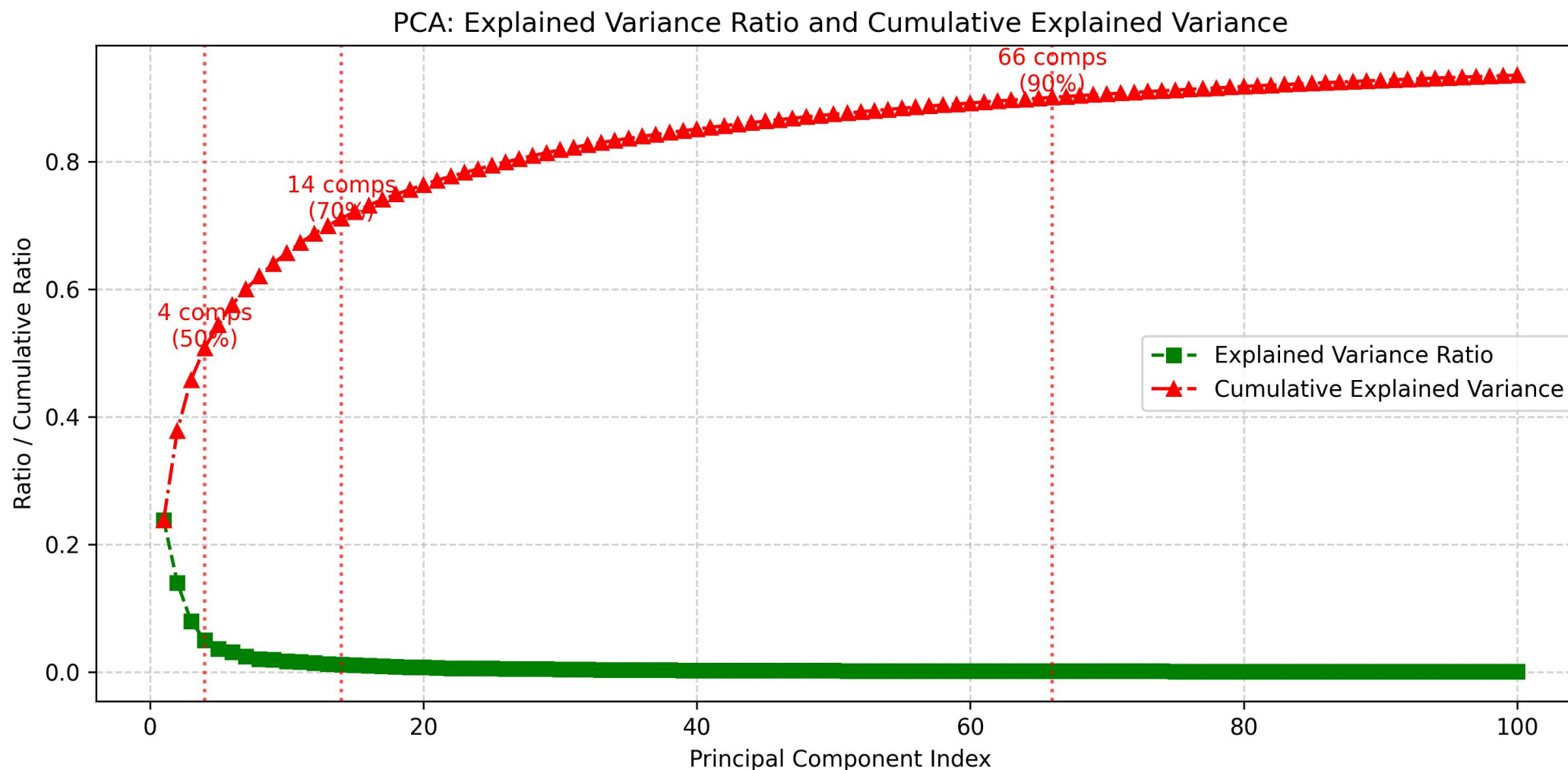
As K increases from 5 to 100, MSE decreases from 0.009 to 0.001 (about 88%), and PSNR rises from approximately 20.8 dB to 28.8 dB, indicating a significant improvement in reconstruction quality. The performance stabilizes after K=50, suggesting that the PCA model has reached convergence in face image reconstruction or become dominated by noise.

reconstruction_records.csv

explained variance



explained variance ratio and cumulative explained variance

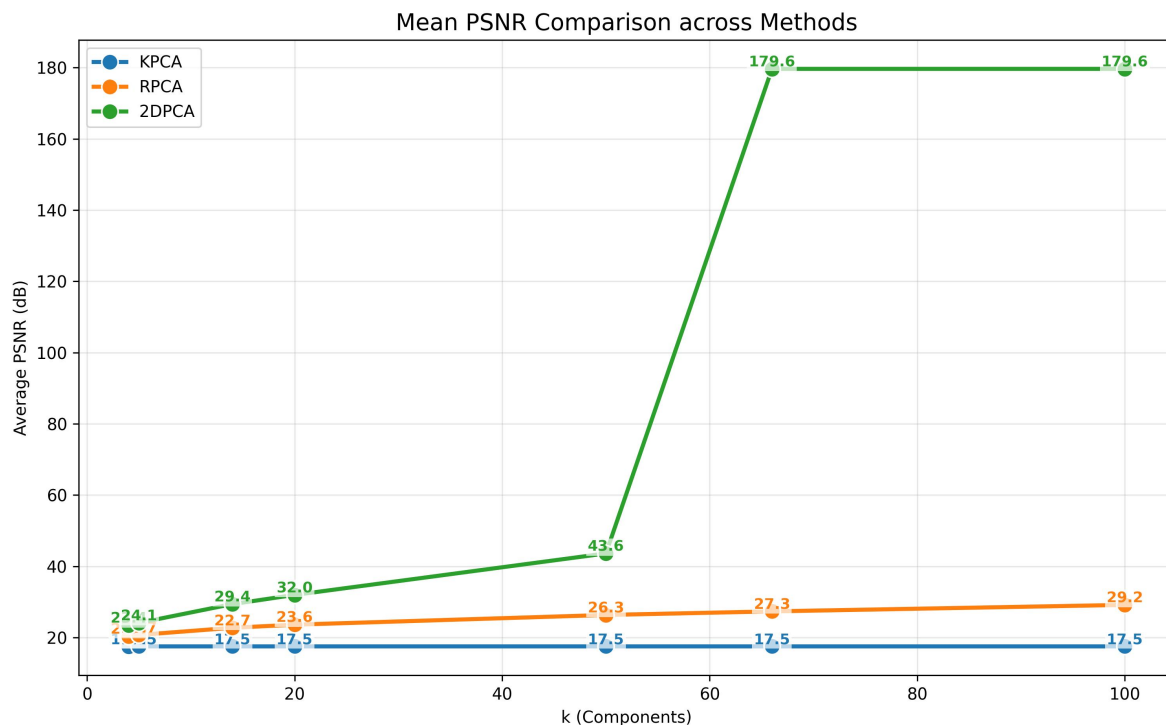


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降维算法流派	极低秩 ($k = 14$)		最优转折点 ($k = 66$)		高维截断 ($k = 100$)	
	PSNR (dB)	Time (s)	PSNR (dB)	Time (s)	PSNR (dB)	Time (s)
Baseline PCA (基线计算)	23.10	29.33	27.36	30.68	29.56	31.02
SVD (纯数值对角加速)	22.80	0.029	27.56	0.051	29.53	0.262
KPCA (非线性核化映射)	17.94	0.169	19.01	0.201	19.26	0.218
RPCA (凸优化稀疏剥离)	22.80	0.286	27.56	0.308	29.53	0.315



SVD (Acceleration): Ultra-fast (0.051s).

The optimal real-time solver.

2DPCA (Topology): Peak PSNR (179.6 dB). 2D structure enables near-lossless recovery.

RPCA (Robustness): Stable baseline performance; perfectly isolates sparse noise.

KPCA (Bottleneck): Fails at 17.5 dB due to the inherent "Pre-image Problem".

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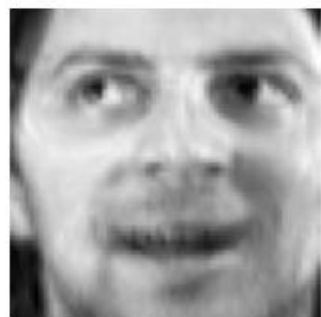


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RPCA (k=100)



PSNR:28.58



PSNR:29.09



PSNR:31.23



PSNR:30.28



PSNR:29.28



PSNR:28.62



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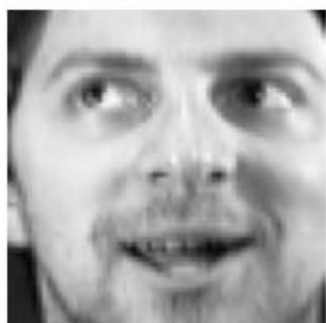


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2DPCA (k=100)



PSNR:178.81



PSNR:173.45



PSNR:182.66



PSNR:174.21



PSNR:179.09



PSNR:174.60

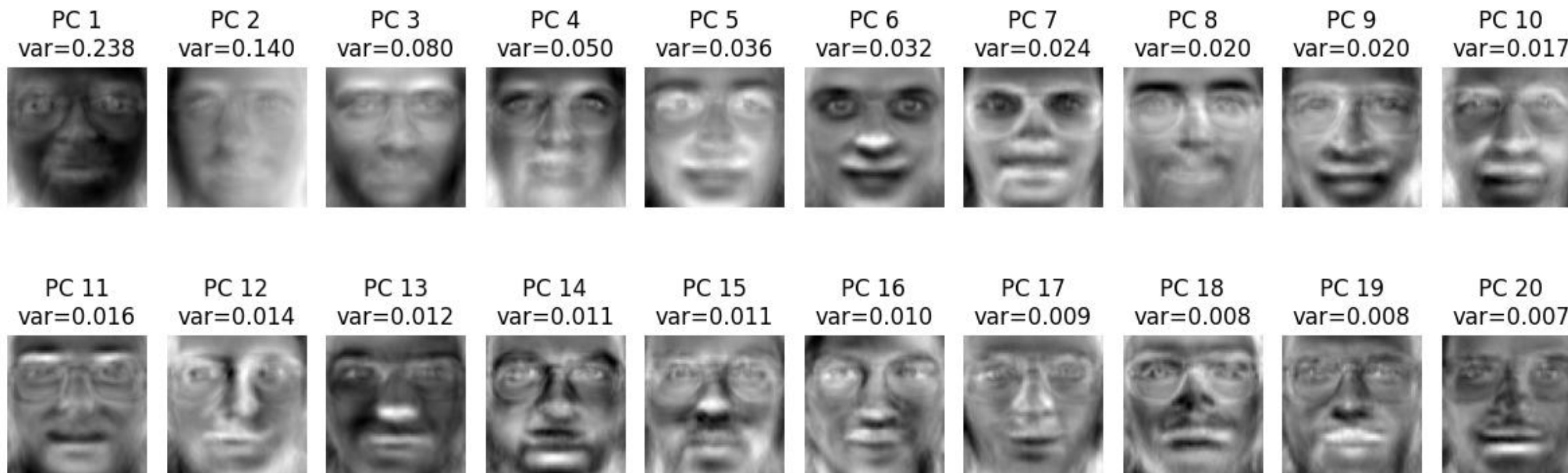


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Top principal components (eigenfaces)



Analysis and Summary

As shown in the tables and figures, this experiment deepened understanding of PCA's principles, procedures, and improvements.

Theoretical insight: SVD-based computation clarified PCA's role in high-dimensional facial feature extraction and dimensionality reduction.

Experimental results: Increasing principal components ($k=5,20,50,100$) improved reconstruction quality, with clearer structures and lower errors.

Performance: For $k \geq 50$, improvements stabilized, indicating major facial features were captured and additional components contributed little.

Improvements: Using explained variance ratio enhanced interpretability, confirming the effectiveness and stability of SVD-based PCA in facial image reconstruction.

References

- [1]Halko, N., Martinsson, P.-G., & Tropp, J. A. (2009). Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2), 217–288.
- [2]Kim, D., & You, K. (2023). PCA, SVD, and Centering of Data. *arXiv preprint*, arXiv:2307.15213.
- [3]Kurian, A. P., Bijitha, S. R., Mohan, L., Kartha, M. M., & Soman, K.P. (2012). Performance Evaluation of Modified SVD based Image Fusion. *International Journal of Computer Applications*, 58(12), 38-45.
- [4]Hu, S., & Du, J. (2022). Research on risk assessment and prediction of RMB internationalization based on the PCA-SA-BPNN model. *Computational Intelligence and Neuroscience*, 2022, Article 5132718.
- [5]Golub, G. H., & Reinsch, C. (1970). Singular value decomposition and least squares solutions. *Numerische Mathematik*, 14, 403–420.



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感谢聆听

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